

Moon Colony : The Next Human Development Frontier

Summary

Establishing a sustainable 100,000-person Moon Colony needs a balanced Earth-Moon transportation system. Based on the defined Space Elevator System, this study builds an integrated optimization framework to evaluate diverse logistics strategies for multiple objectives.

For Problem 1, a **Cost–Time Trade-off Optimization Model** is constructed to characterize the fundamental conflict between transportation cost and project duration under ideal operating conditions. The **Pareto frontier** is generated through a **single-variable discretized exhaustive search** over the transportation allocation decision, enabling precise identification of the **knee point** on the frontier while ensuring global optimality and algorithmic transparency. obtained from the model assigns **74.8%** of total materials to the Space Elevator System, completing construction in **139.2 years** at a total cost of **271.1 trillion USD**, representing a substantial cost reduction relative to the minimum-time strategy with a moderate increase in project duration.

For Problem 2, a **Robust Cost–Time Trade-off Optimization Model** is developed by incorporating non-ideal operational factors, including Coriolis-force-induced tether constraints, tidal-force-related elevator downtime, and rocket launch failures. These physical and reliability constraints are embedded directly into system capacity estimation, yielding a revised Pareto frontier under realistic uncertainty. The robust knee-point strategy allocates **72.35%** of transportation to the space elevator, with a construction period of **164.52 years** and a total cost of **296.33 trillion USD**. Compared with the ideal-condition solution, the project duration increases by approximately **18.2%** and total cost rises by about **9.3%**, reflecting the impact of reliability constraints while preserving the dominant role of the space elevator.

For Problem 3, long-term water logistics are formulated as a finite-horizon, multi-stage decision problem. A **Dynamic Water Supply Optimization Model**, solved using a **stage-wise optimization framework**, minimizes annual transportation cost while strictly satisfying survival-oriented inventory constraints, ensuring uninterrupted life support after colony establishment. The resulting optimal policy specifies discrete replenishment cycles with a total annual water transportation budget of **500.29 million credits**, requiring **17 transportation missions** to maintain safe inventory levels.

For Problem 4, environmental consequences are explicitly incorporated through an **Environmental Impact–Oriented Decision Model**, which introduces an **Environmental Impact Coefficient** based on radiative forcing equivalence to compare emissions from different transportation strategies. The optimal strategy assigns **88.17%** of materials to the Space Elevator **System**, achieving completion in **89.12 years** with a total cost of **141.05 trillion USD**, further reinforcing the advantage of space-elevator-dominated logistics.

Finally, for Problem 5, we provide a recommendation letter summarizing the optimal strategy for cost, time, reliability, and environmental objectives.

Keywords: Pareto frontier; single-variable discretized exhaustive search; knee-point; stage-wise optimization framework; Environmental Impact Coefficient

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1 Introduction

1.1 Problem Background

As space technology advances, human activities are moving from short-term missions toward long-term habitation. The MCM vision proposes a Moon Colony supporting 100,000 inhabitants by 2050, requiring the transport of approximately 100 million metric tons of construction materials from Earth to the Moon, followed by sustained resupply of water and life-support resources. This unprecedented demand challenges existing Earth–Moon transportation systems.

Conventional chemical-propulsion rockets are technologically mature but face high marginal costs and limited launch capacity at large scales. In contrast, the space elevator system, powered by electricity, offers a scalable infrastructure for interplanetary logistics, commerce, and exploration, with potential advantages in cost efficiency and environmental impact.

Given the complexity and uncertainty of lunar logistics, this study develops a multi-objective modeling framework to evaluate rocket-based, space-elevator-based, and hybrid transportation strategies for sustainable Moon Colony development.

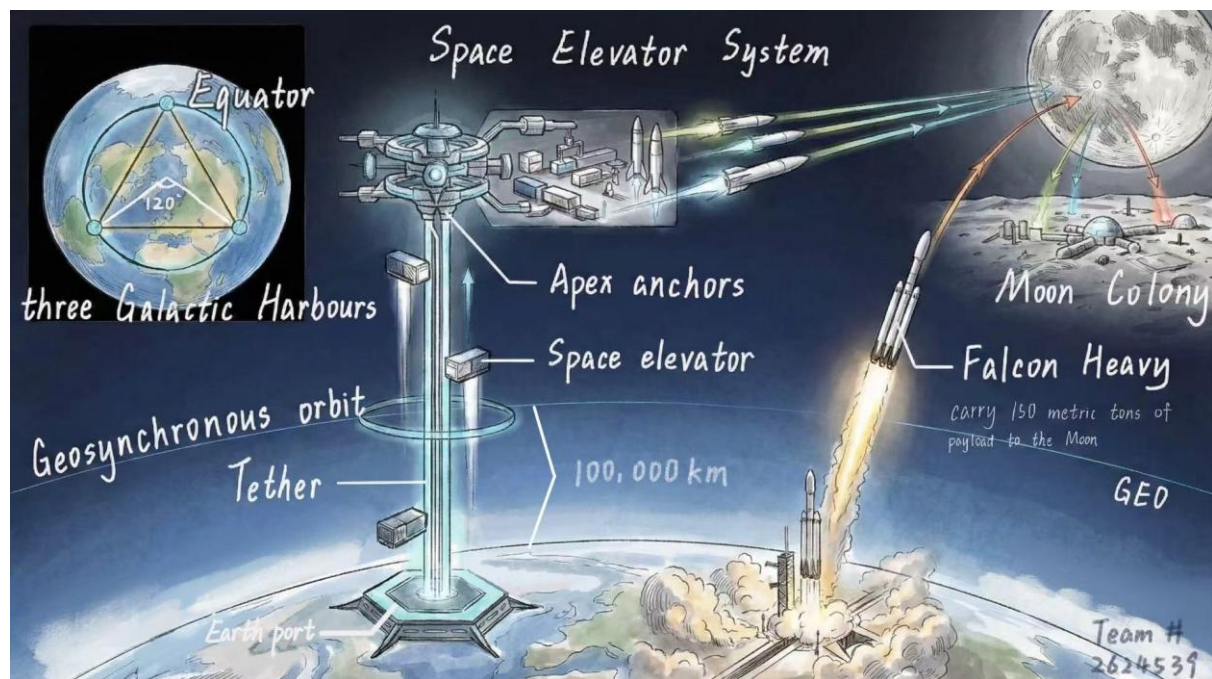


Figure 1: Space Elevator System vs Traditional Rocket Launches

1.2 Restatement of the Problem

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

- **Problem 1** For the transportation of 100 million metric tons of construction materials required for building a Moon Colony, develop comparative models under ideal conditions for transporting materials from Earth to the Moon using three different strategies:
 - (a) Using the Space Elevator System's three Galactic Harbor's alone;

- (b) Traditional rocket launches via the Falcon Heavy alone;
- (c) some combination of the two methods.
- **Problem 2** Analyze the impact of non-ideal factors—including tether swaying, rocket launch failures, and space elevator breakdowns—on the transportation strategies developed in Problem 1, and assess how these uncertainties affect the cost and associated timeline.
- **Problem 3** Develop a Moon Colony Water Logistics Model to estimate the annual water supply and transportation.
- **Problem 4** Assess the environmental impacts on Earth associated with each transportation scenario, considering factors such as energy consumption, and emissions, and propose strategies to minimize the overall environmental footprint of space transportation.
- **Problem 5** Based on the analyses conducted in the previous problems, recommend an optimal strategy for building and sustaining a 100,000-person Moon Colony, balancing cost efficiency, reliability, environmental impact, and long-term sustainability.

1.3 Literature Review

As human space exploration transitions from short-duration missions toward large-scale settlement, the establishment of a Moon Colony accommodating 100,000 inhabitants poses significant challenges to Earth–Moon logistics systems. Although commercial space companies represented by SpaceX have substantially reduced launch costs through rocket reusability technologies, the relatively low payload efficiency of chemical rockets, along with carbon dioxide and soot emissions released in the stratosphere, remain critical limitations. In contrast, the space elevator is widely regarded as a sustainable “green pathway” for space transportation. By utilizing ultra-high-strength materials such as monocrystalline graphene to construct a tether system extending up to 100,000 kilometers, space elevators are expected to increase payload efficiency to approximately 70%, significantly improving the economic feasibility of large-scale material transport.

From the perspective of physical and mechanical stability, the tether system is subject to complex influences from Coriolis forces and tidal forces. As an inertial force in a rotating reference frame, the Coriolis force induces westward deflection and oscillations of the tether as climbers ascend, requiring precise dynamic compensation and control algorithms. Meanwhile, tidal forces arising from gravitational gradients act as a natural stabilizing mechanism, maintaining tension balance inside and outside geostationary orbit (GEO) and preserving the vertical configuration of the tether structure.

The long-term sustainability of a lunar colony critically depends on the efficiency of closed-loop life-support systems, with water resource management as a central component. Based on operational data from the International Space Station (ISS), current systems can achieve an overall water recovery rate of approximately 92%, with future upgrades targeting recovery rates as high as 99%. Such high-efficiency recycling systems, combined with the low-energy transportation advantages of space elevators, form the technological and economic foundation for the feasibility of large-scale lunar settlement.

1.4 Our Work

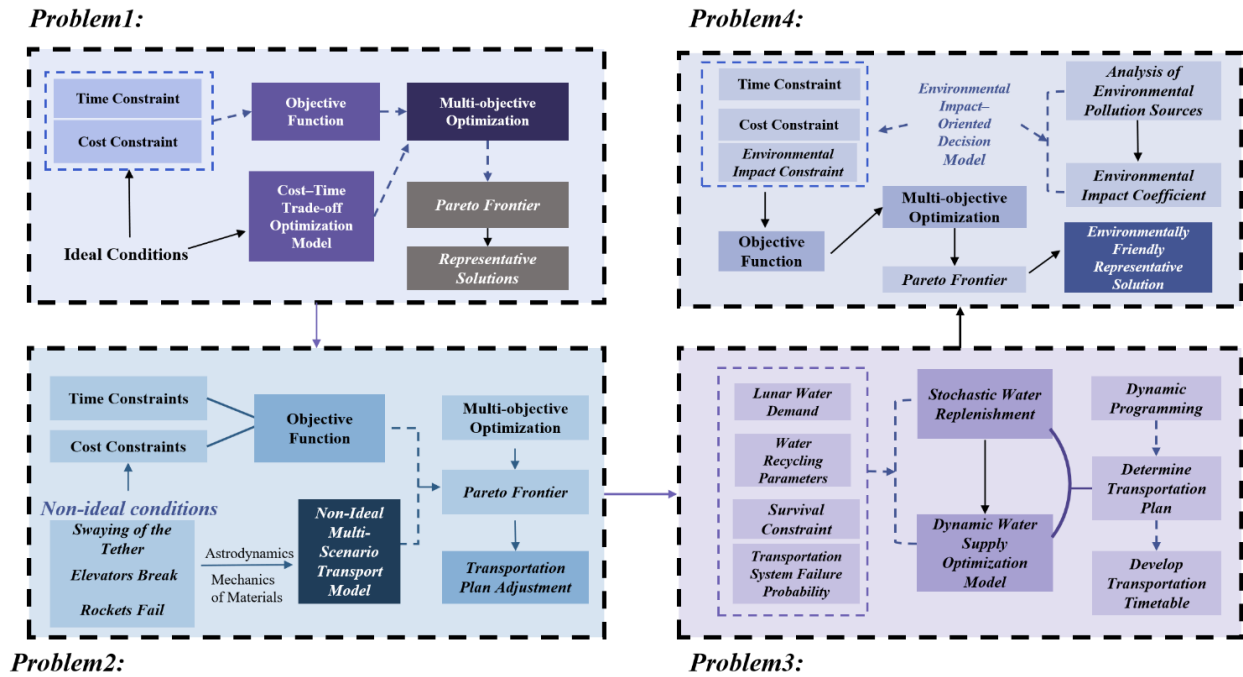


Figure 2: Our work

2 Assumptions and Justifications

★H1. Ideal Logistics Capacity and Material Equivalence.

Under ideal operating conditions, we neglect safety margins, redundancy design, micro-meteoroid impacts, and unexpected maintenance downtime. All transportation tasks are assumed to operate without empty or partially loaded payloads, and all materials are treated as equivalent in transportation attributes such as volume, physical form, and packaging. The annual transport capacity of a Galactic Harbour is set to 179,000 tons, and the payload of each rocket launch is assumed to be 150 tons.^[1]

Justification: To support migration at the scale of 100,000 inhabitants, the logistics system must rely on highly standardized and modularized transport units. This assumption establishes a deterministic baseline, enabling the model to focus on macroscopic throughput efficiency rather than micro-level loading randomness.

★H2. High-Frequency Turnaround of Rocket Launch Sites.

We assume that ten major rocket launch sites worldwide can support Falcon Heavy-class launch and recovery operations with a 24-hour turnaround cycle. The Moon Colony construction is treated as a globally coordinated mission, with no political or institutional constraints limiting launch capacity.

Justification: Large-scale lunar construction requires globally integrated infrastructure. In the technological context of 2050, airline-style rocket reusability and rapid turnaround represent a necessary target, allowing the system to be modeled as a continuous launch flow rather than discrete events.^[11]

★H3. System Parallelism and Resource Independence.

The Space Elevator System and rocket transportation system are assumed to operate stably with fully parallel subsystems. No competition for shared resources such as energy, personnel, or communication bandwidth is considered; thus, total throughput equals the linear sum of independent subsystem capacities.

Justification: As a global-scale engineering project, sufficient redundancy would be designed to ensure operational independence, allowing the model to focus on evaluating transportation potential rather than resource coupling.

★H4. Linear Transportation Cost.

Total transportation cost is assumed to be a linear function of transported mass, excluding economies of scale or dynamic pricing effects.

Justification: Linear cost functions ensure global optimality in multi-objective optimization and are reasonable under long-term, government-led infrastructure contracts.

★H5. Ideal Physics and Constant-Speed Ascent.

Air resistance is neglected, and space elevator climbers are assumed to operate at constant speed, ignoring acceleration and deceleration phases.

Justification: Given the 100,000 km tether length, most of the system lies beyond the atmosphere, making aerodynamic effects negligible and enabling a simple, deterministic physical baseline.

★H6. Negligible Environmental Impact of Mass Transfer.

The removal of 100 million tons of material from Earth is assumed to have no direct impact on Earth's orbital dynamics or ecological balance; environmental assessment focuses solely on emissions, energy use, and space debris.^[11]

Justification: This mass represents approximately 1.67×10^{-14} of Earth's total mass and is physically negligible.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description
Cap_{GH}	Ideal annual transportation capacity of the Galactic Harbor
N_{GH}	Number of Galactic Harbour
N_{bases}	Number of rocket launch site
F_{freq}	Annual launch frequency per rocket launch site
$M_{payload}$	Payload capacity per launch of Falcon Heavy

4 Cost–Time Trade-off Optimization Model

To assess the feasibility of large-scale material transportation during the construction phase of the Moon Colony, this study develops four Cost–Time Trade-off Optimization Models that evaluate transportation time and cost as core performance metrics, systematically comparing three Earth–Moon transport strategies for a total of 1×10^8 tons of construction materials.

4.1 Data Description

Due to the lack of operational data for large-scale Space Elevator System, this study adopts representative cost estimates from authoritative literature. For the space elevator system, the unit transportation cost is adopted from Edwards (2007), which estimates the operational cost of a mature space elevator (P_{SE}) at approximately \$250 per kilogram. This estimate has been widely cited in subsequent space transportation studies and is considered representative of a fully developed system. For traditional rocket launches, commercial estimates for modern launch systems indicate costs (P_{Rocket}) on the order of \$10,000 per kilogram.^{[2][3]}

4.2 The Establishment of CTTO Model

The total annual capacity of the Space Elevator System is therefore given by

$$R_{SE} = N_{GH} \times Cap_{GH} \quad (1)$$

The total annual capacity for the rocket transportation system of

$$R_{rocket} = N_{bases} \times F_{freq} \times M_{payload} \quad (2)$$

We define the decision variable $\alpha \in [0, 1]$ as the proportion of total material demand undertaken by the Space Elevator System. The transportation volume carried by the Space Elevator System is given by

$$Q_{SE} = \alpha \cdot D_{total} \quad (3)$$

While the transportation volume carried by the traditional rocket system is

$$Q_{Rocket} = (1 - \alpha) \cdot D_{total} \quad (4)$$

When $\alpha = 1$, the model reduces to Plan a, in which transportation relies exclusively on the Space Elevator System composed of three Galactic Harbours. When $\alpha = 0$, the model corresponds to Plan b, representing transportation solely via traditional rocket launches from existing bases. All intermediate values of α constitute Plan c, which represents a hybrid strategy combining both transportation methods.

Based on these allocations, the total transportation cost can be expressed as

$$Z_1(\alpha) = 1000 \times (Q_{SE} \cdot P_{SE} + Q_{Rocket} \cdot P_{Rocket}) \quad (5)$$

Where all unit costs are uniformly converted to USD per ton.

The time required for the Space Elevator System to complete its assigned transportation task is denoted as

$$T_{SE} = \frac{Q_{SE}}{R_{SE}} \quad (6)$$

While the time required for the traditional rocket system is

$$T_{Rocket} = \frac{Q_{Rocket}}{R_{Rocket}} \quad (7)$$

Since the two transportation systems operate in parallel, the overall completion time of the transportation task is determined by the slower system

$$Z_2(\alpha) = \max(T_{SE}, T_{Rocket}) \quad (8)$$

4.3 The Solution of CTTO Model

In a multi-objective optimization problem, all solutions on the Pareto frontier are mathematically non-dominated. However, not all Pareto-optimal solutions are equally practical from

an engineering perspective. It is therefore necessary to further identify a representative solution on the Pareto frontier that achieves a well-balanced trade-off among multiple objectives, commonly referred to as the knee point.

To this end, this study introduces the concept of the Utopia Point as a reference benchmark. The utopia point is defined as the hypothetical solution that simultaneously attains the theoretical optimum of each objective, namely

$$Utopia\ Point = (Z_1^{\min}, Z_2^{\min}) \quad (9)$$

Where $Z_1^{\min}$ denotes the minimum transportation cost on the Pareto frontier and $Z_2^{\min}$ represents the shortest project completion time.

Because transportation cost and project duration differ substantially in both units and magnitude, directly computing their geometric distance would cause the result to be dominated by a single objective. To address this issue, we apply Min–Max normalization to the objective values along the Pareto frontier, mapping them onto a dimensionless interval $[0, 1]$:

$$Z_i^{norm} = \frac{Z_i - Z_i^{\min}}{Z_i^{\max} - Z_i^{\min}}, \quad i = 1, 2 \quad (10)$$

The normalized utopia point can therefore be expressed as $(0, 0)$, implying that the two objectives are assigned equal weight in the evaluation process.

In the normalized objective space, we adopt the Euclidean distance to measure the deviation of each solution from the utopia point:

$$d(\alpha) = \sqrt{(Z_1^{norm}(\alpha))^2 + (Z_2^{norm}(\alpha))^2} \quad (11)$$

This distance reflects the overall performance loss of a given solution in terms of its simultaneous deviation from both the minimum transportation cost and the shortest project duration. A smaller distance indicates that the solution is closer to the ideal state across the two objectives.

By searching along the Pareto frontier for the solution that minimizes $d(\alpha)$

$$\alpha^* = \arg \min_{\alpha \in [0, 1]} d(\alpha) \quad (12)$$

The corresponding knee point solution can be identified.

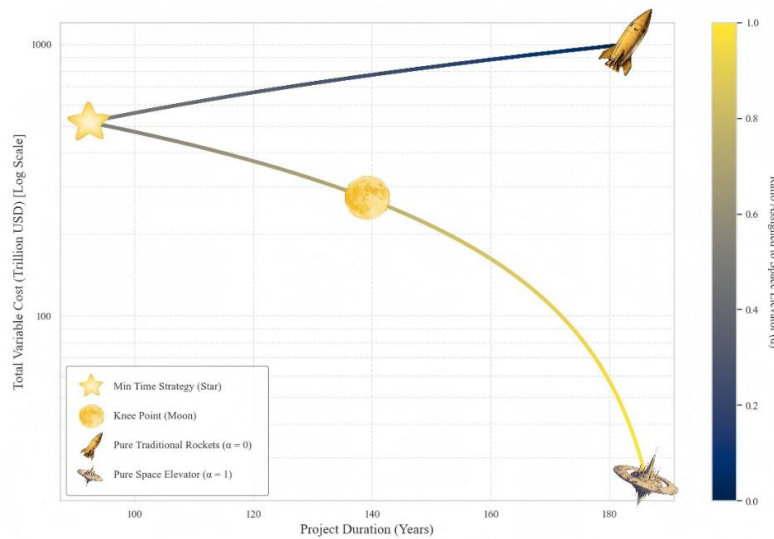


Figure 3: Distribution of the Pareto Frontier in the CTTO Model

Table 2: Four representative strategies along the Pareto frontier

Strategy Name	Time (Years)	Cost (Trillion)	α (SE%)
Pure Space Elevator	186.2	25.0	100.0%
Pure Rockets	182.6	1000.0	0.0%
Min Time Strategy	92.2	517.3	49.5%
Knee Point	139.2	271.1	74.8%

The solution identifies four representative strategies along the Pareto frontier: pure rocket transportation, pure space elevator transportation, the minimum-time strategy, and the knee point strategy. The pure rocket strategy requires 186.2 years and incurs a total cost of \$1000.0 trillion, showing extremely poor economic performance. The pure space elevator strategy minimizes cost to \$25.0 trillion but extends the project duration to 186.2 years. The minimum-time strategy completes the task in 92.2 years with a cost of \$517.3 trillion by assigning 49.5% of transport to the space elevator system. The knee point strategy, with 74.8% allocated to space elevators, completes the mission in 139.2 years at a cost of \$271.1 trillion. Compared with the minimum-time strategy, it saves \$246.1 trillion while increasing duration by 47.0 years. The logarithmic cost scale highlights this trade-off, confirming the knee point as the most practical solution.

5 Robust Cost–Time Trade-off Optimization Model

5.1 Data Description

In order to quantify the impact of rocket launch failures in Model 2, historical launch reliability data of SpaceX launch vehicles are incorporated. According to publicly available statistics, the Falcon 9 launch vehicle has achieved a success rate of 99.38%, with 480 successful missions out of 483 launches. Falcon Heavy, which is selected in this study as the representative model for traditional rocket-based transportation, has recorded a 100% success rate across 11 launches to date.^{[6][8][11]}

5.2 The Establishment of RCTTO Model

In Problem 1, we analyzed the transportation time and cost associated with three different strategies under ideal operating conditions. However, real-world engineering systems rarely function in a perfectly ideal manner. Therefore, building upon the results of Problem 1, we incorporate several non-ideal operational factors and quantitatively evaluate their impacts on transportation time and cost, thereby deriving a revised optimal transportation strategy under realistic conditions.

5.2.1 Imperfect Operational Factors of the Space Elevator System

• Impact of Tether Swaying

Under ideal conditions, the tether of the Space Elevator System is assumed to remain perfectly vertical without any lateral motion. In reality, Earth is a rotating reference frame, and any object moving relative to Earth is subject to the Coriolis force. The core components of the

Space Elevator System consist of a tether extending from Earth's surface to beyond geosynchronous orbit (GEO) and climbers that travel along the tether. As the Earth rotates, the tether rotates together with it. When a climber ascends or descends along the tether, its motion relative to the rotating Earth introduces a velocity component that gives rise to the Coriolis force, which acts on both the climber and the attached tether.^[4]

$$\vec{F}_c = -2m(\vec{\omega} \times \vec{v}) \quad (13)$$

Where m is the mass of the climber, $\vec{\omega}$ is Earth's angular velocity vector, and $\vec{v}$ is the climber's velocity relative to the rotating reference frame.

Because the climbing direction is perpendicular to Earth's rotation axis, the magnitude of the Coriolis force can be written as

$$F_c = 2m\omega v \quad (14)$$

Under the action of the Coriolis force, the tether undergoes lateral deformation and deviates from the vertical direction. Let θ denote the deflection (lag) angle of the tether, and let T denote the total tether tension. The horizontal component of the tether tension is given by

$$F_h = T \sin \theta \quad (15)$$

Under the assumption of constant-speed ascent, the net lateral force acting on the system is zero. Neglecting other external forces, force balance yields

$$F_h = F_c \quad (16)$$

Since the Coriolis force is proportional to the climbing speed, higher ascent velocities lead to larger Coriolis forces and thus higher tether tension. However, the total tension that the tether can sustain is strictly limited by the tensile strength of the material. Let $T_{\max}$ denote the maximum allowable tension determined by the permissible design stress $\sigma_{\max}$ of the graphene material. To ensure structural stability, neither the tether tension nor the deflection angle can increase without bound. Consequently, under combined material strength and geometric stability constraints, the above equilibrium relationship can be inverted to determine the maximum allowable quasi-steady climbing speed of the system.^[5]

The relationship between the maximum tether tension and the allowable design stress of graphene is given by

$$T_{\max} = \sigma_{\max} \cdot A \quad (17)$$

Where A is the cross-sectional area of the tether.

Let M denote the total mass of a single tether. For a 100,000 km-long graphene tether, the average cross-sectional area can be expressed as

$$\bar{A} = \frac{M}{\rho \cdot L} \quad (18)$$

Where ρ is the density of graphene and L is the total tether length.

By jointly solving Eqs(14)–(18), the maximum allowable quasi-steady climbing speed $v_{\max}$ of the Space Elevator System can be obtained as

$$v_{\max} = \frac{\sigma_{\max} M \sin \theta}{2\rho L m w} \quad (19)$$

• Effect of Elevator Breakdowns

This study adopts annual downtime (in days) as the primary indicator for quantifying the operational impact of failures in the Space Elevator System. This metric enables engineering-

level reliability considerations to be directly mapped onto system-level annual availability, thereby establishing a clear linkage with key operational parameters such as transportation capacity, economic performance, and investment payback period. As a result, it provides an effective measure of the practical feasibility of the space elevator as a high-frequency, low-cost space transportation infrastructure.^[10]

The periodicity of tidal forces constitutes a major contributor to elevator downtime. Tidal effects influence operational safety and efficiency primarily by inducing dynamic responses in the tether, including oscillations and lateral swaying. These forces arise from gravitational field non-uniformity, whereby the magnitude and direction of gravitational attraction vary across different parts of an extended structure. For the Space Elevator System, the dominant sources of tidal forces are the gravitational interactions with the Moon and the Sun acting on the Earth and the extended tether.

Based on the above considerations, we assume that during each spring tide period, when tidal forces reach their peak, the space elevator cannot maintain safe and efficient operation for certain portions of each day. Estimating that approximately 4–6 days per month are subject to significant tidal disturbances, the cumulative annual duration during which elevator operations are constrained is approximately 48–72 days.^{[6][7]}

It should be noted that this estimation is derived under conservative assumptions that do not fully account for future technological advancements. Given that the time horizon of this study extends to 2050 and beyond, it is reasonable to assume that material performance, structural control technologies, and operational scheduling strategies related to the Space Elevator System will have reached a relatively mature stage, thereby partially mitigating the adverse effects of tidal forces.

Therefore, to maintain conservatism while avoiding excessive underestimation of long-term system capacity, we adopt the lower bound of 48 days as the representative annual downtime due to significant tidal effects. This parameter is used consistently in subsequent model calculations and performance evaluations. Accordingly, the relationship between the effective annual transportation capacity and the ideal annual capacity of the space elevator can be expressed as

$$\widetilde{C}_e = \frac{365 - 48}{365} C_e = \frac{317}{365} C_e \quad (20)$$

• Effect of Elevator Breakdowns

Rocket launches are inherently subject to dynamic and environmental uncertainties, and thus launch success rates are typically lower than 100%.

Due to the limited number of Falcon Heavy launches, its observed success rate may not yet be statistically representative. Considering that Falcon Heavy and Falcon 9 are developed by the same company, share core technologies, and belong to the same launch vehicle family, it is reasonable to assume comparable reliability characteristics between the two systems. Therefore, to obtain a more conservative and statistically stable estimate, the success rate of Falcon 9 is adopted as a proxy for the launch success probability of Falcon Heavy in this study.

5.3 The Solution of RCTTO Model

Under continuous operation, the one-way operation time of a Galactic Harbour is given

by

$$t_{Galactic\ Harbor} = \frac{L}{v_{\max}} = \frac{2\rho L^2 m \omega}{\sigma_{\max} M \sin \theta} \quad (21)$$

Given an annual available operating time of $t_{year} = 317 \times 24 \times 3600$, the number of feasible transportation cycles per year can be expressed as

$$n_{year} = \frac{t_{year}}{t_{Galactic\ Harbor}} = \frac{t_{year} \sigma_{\max} M \sin \theta}{2\rho L^2 m \omega} \quad (22)$$

Accordingly, the annual transportation capacity of a Galactic Harbour is

$$Cap_{year} = n_{year} \cdot m = \frac{t_{year} \sigma_{\max} M \sin \theta}{2\rho L^2 \omega} \quad (23)$$

Where m denotes the payload mass per operation.

Notably, the payload mass m is eliminated in the final expression when constraints imposed by Coriolis-force-limited climbing speed are applied. This result indicates that, under Coriolis force constraints, the annual transportation capacity of a Galactic Harbour becomes independent of the single-trip payload mass, and is instead governed by physical and temporal limitations of the system.

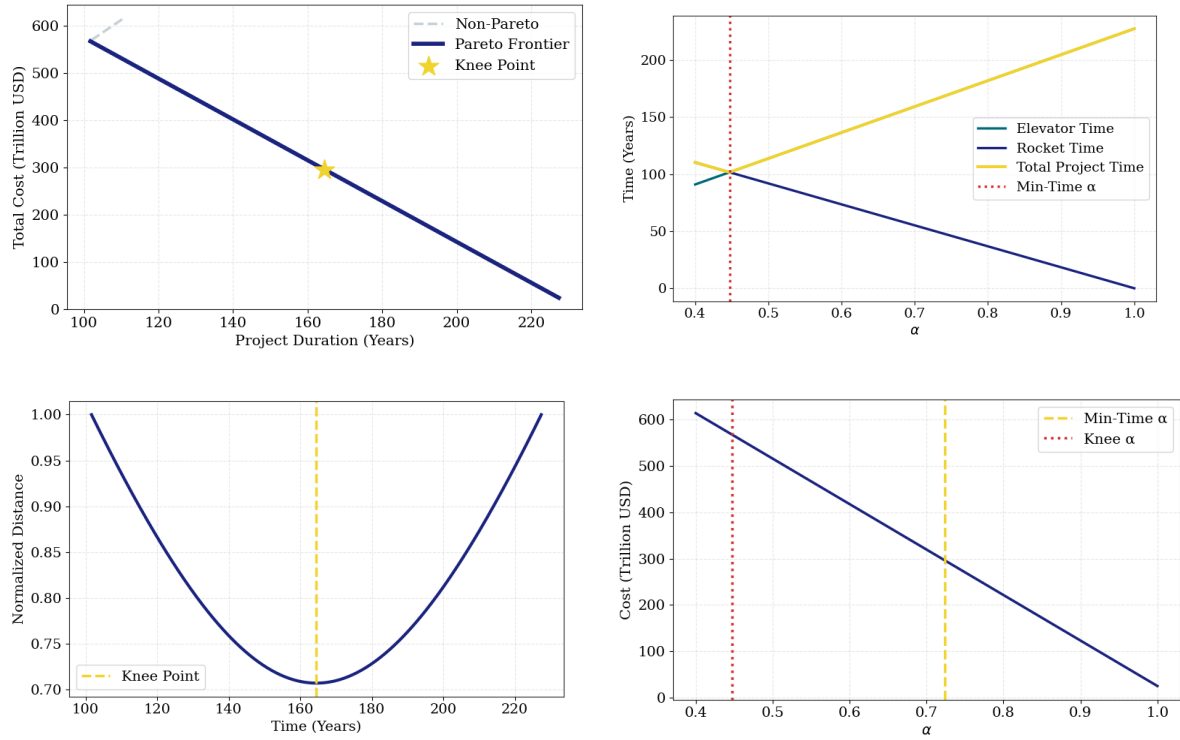


Figure 4: Distribution of the Pareto Frontier in the RCTTO Model

Table 3: Three representative transportation strategies

Strategy Name	Time (Years)	Cost (Trillion)	α (SE%)
Pure Space Elevator	227.40	25.0	100.0%
Min Time Strategy	101.64	567.66	44.7%
Knee Point	164.52	296.33	70.71%

Based on Pareto-optimal analysis, three representative transportation strategies are identified under non-ideal conditions. The minimum-time strategy achieves the shortest project duration of 101.64 years by allocating 44.70% of transportation to the space elevator system; however, this strategy incurs a very high total cost of \$567.66 trillion. The baseline strategy, which relies entirely on the space elevator, minimizes total cost at \$25.00 trillion, but requires an extended completion time of 227.40 years, making it impractical for timely lunar development.

Applying a knee point identification method on the Pareto frontier yields an optimal balance strategy. With 72.35% of materials transported via the space elevator, this strategy completes the project in 164.52 years at a total cost of \$296.33 trillion, and attains the minimum distance to the utopia point, indicating a robust trade-off between transportation time and cost under system uncertainties.

6 Dynamic Water Supply Optimization Model

6.1 Data Description

International Space Station (ISS) life support systems currently achieve water recovery up to 98% through advanced recycling of urine, sweat, and humidity condensate, compared with earlier rates of 93–94% (NASA reports). Daily water requirement per person is taken as 4.4 L based on spaceflight life-support studies. Given future technology and long-term lunar habitation, a 99% recycling rate is assumed for the Moon Colony model.^[9]

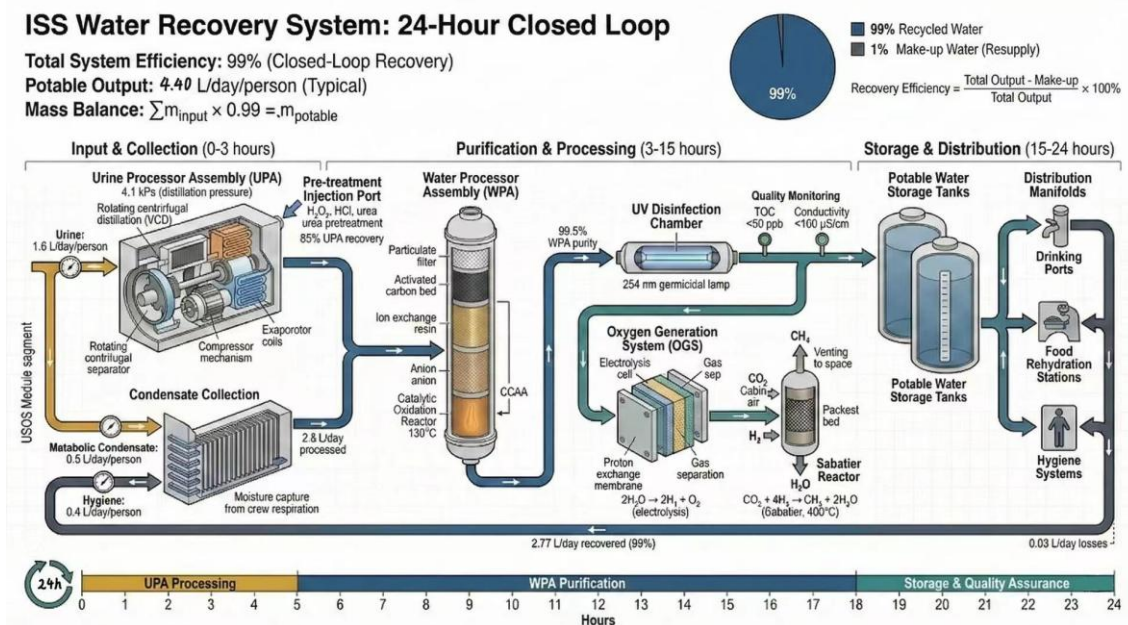


Figure 5:ISS Water Recovery System

6.2 The Establishment of DWSO Model

Problem3 can be formulated as a finite-horizon, multi-stage decision process, whose objective is to minimize total annual transportation cost subject to inventory constraints. The system is modeled as a dynamic process jointly governed by inventory state evolution and the cost structures of multiple transportation modes, and is solved using dynamic programming.

To ensure model tractability and interpretability, we assume that the population size remains constant throughout the study period and that survival risk arises solely from insufficient water supply. No other exogenous mortality factors are introduced.

Water loss is assumed to occur during the daily supply cycle. According to the data description, the water recycling rate is 99%.

Let I_{t-1} denote the inventory level at the end of day $t-1$. At the beginning of day t , a shipment of water arrives instantaneously, after which daily consumption and recycling take place. The daily water loss L is given by

$$L = S_{\min} \cdot (1 - \eta) \quad (24)$$

Where $S_{\min}$ is minimum daily water requirement for the Moon Colony.

Accordingly, the inventory level at the end of day t satisfies the following difference equation:

$$I_t = I_{t-1} - L + Q_t \quad (25)$$

• Constraint 1: Survival Constraint

Existing studies and the problem specification indicate that individuals face mortality risk if their daily water intake falls below 4.4 L for three consecutive days. However, for humanitarian considerations and system robustness, we do not exploit this threshold as a binding condition to compress transportation cost.

Instead, a more conservative and robust constraint is imposed: the water inventory must never fall below a safe inventory level at any time, ensuring that no individual enters a potential survival risk state. Formally,

$$I_t \geq S_{\min}, \forall t \in \{1, 2, \dots, T\} \quad (26)$$

• Constraint 2: Boundary Conditions

At the initial state ($t = 1, \text{January } 1, 2220$), the inventory level prior to replenishment is assumed to be exactly at the safety threshold:

$$I_0 = S_{\min} \quad (27)$$

At the terminal state ($\text{December } 31, 2220$), the inventory level must also satisfy

$$I_T \geq S_{\min} \quad (28)$$

• Constraint 3: Transportation System Failure Probability

From Eq. (20), the daily success probability of the Space Elevator System p_{SE} is $1 - \frac{48}{365}$, while the success probability of rocket transportation p_R is 0.9938. If a transportation attempt fails, the corresponding payload is completely lost for that day.

6.3 The Solution of DWSO Model

To apply dynamic programming, we first determine the minimum replenishment quantity required on day t to sustain water supply for k consecutive days.

Under natural decay without intermediate replenishment, the inventory evolution over k days satisfies

$$(I_{t-1} + Q) - k \cdot L \geq S_{\min} \quad (29)$$

The supply quantity $D(k)$ on day t :

$$D(k) \geq k \cdot L \quad (30)$$

Given a fixed replenishment period k , the problem reduces to determining the optimal combination of transportation modes that minimizes the cost of delivering quantity $Q = D(k)$.

Let $i = 1$ denote the Space Elevator System and $i = 2$ denote traditional rocket transportation. Define the cost function $W(Q)$ as the minimum cost required to transport total quantity Q :

$$W(Q) = \min_{x_1, x_2, v_1, v_2} \sum_{i=1}^2 (F_i x_i + C_i v_i) \quad (31)$$

Where x_i denotes the number of uses of transportation mode i , and C_i and v_i represent the corresponding unit cost and transported volume.

The optimization is subject to the constraint

$$\begin{aligned} v_1 + v_2 &\geq Q \\ v_i &\leq x_i \cdot M_i, i \in \{1, 2\} \\ x_i &\in N_+, v_i \geq 0 \end{aligned} \quad (32)$$

Because only two transportation modes are available and the maximum payload of rockets is significantly smaller than that of the Space Elevator System ($M_2 \ll M_1$), this problem can be efficiently solved by enumerating the number of space elevator uses. Specifically, x_1 is enumerated over the interval $\left[0, \left\lceil \frac{Q}{M_1} \right\rceil\right]$. For each feasible x_1 , the remaining transportation demand

$$\Delta Q = \max\{0, Q - x_1 \cdot M_1\} \quad (33)$$

Yielding the corresponding number of rocket launches

$$x_2 = \left\lceil \frac{\Delta Q}{M_2} \right\rceil \quad (34)$$

By evaluating total cost across all feasible values of x_1 , the minimum-cost solution is selected as $W(Q)$.

Assuming the construction of a Moon Colony with an estimated population of 100,000 beginning in the year 2050, and adopting the construction duration of 164.52 years obtained in Question 2, we illustrate the water transportation schedule using the year 2220 as a representative example. The resulting transportation timetable is summarized in Table 4.

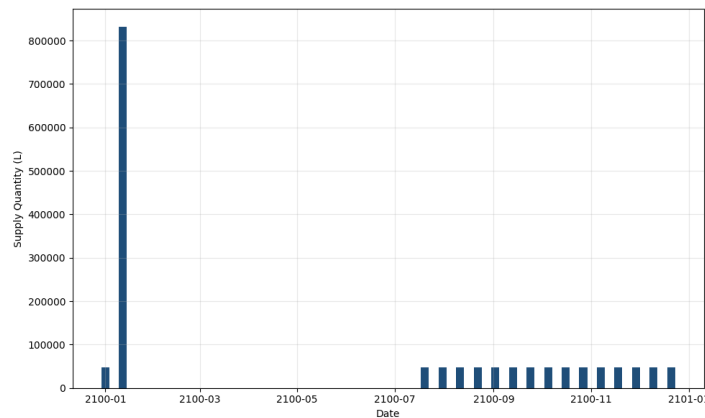


Figure 6: The resulting transportation timetable

Table 4: The resulting transportation timetable

Supply Date	Interval (Days)	Quantity (L)	SE Trips	Rocket Trips	Cost (Million Credits)
2220-01-01	11	48400	1	0	15.0836
2220-01-12	189	831600	17	0	258.9543
2220-07-20	11	48400	1	0	15.0836
2220-07-31	11	48400	1	0	15.0836
2220-08-11	11	48400	1	0	15.0836
2220-08-22	11	48400	1	0	15.0836
2220-09-02	11	48400	1	0	15.0836
2220-09-13	11	48400	1	0	15.0836
2220-09-24	11	48400	1	0	15.0836
2220-10-05	11	48400	1	0	15.0836
2220-10-16	11	48400	1	0	15.0836
2220-10-27	11	48400	1	0	15.0836
2220-11-07	11	48400	1	0	15.0836
2220-11-18	11	48400	1	0	15.0836
2220-11-29	11	48400	1	0	15.0836
2220-12-10	11	48400	1	0	15.0836
2220-12-21	11	48400	1	0	15.0836

The total annual budget is 500.2918 million credits, and a total of 17 transportation missions are required over the year.

7 Environmental Impact–Oriented Decision Model

7.1 Data Description

To quantify environmental impacts, this study adopts an Environmental Impact Coefficient (EIC) based on radiative forcing equivalence. Rocket emission parameters are based on Falcon Heavy data, including $M_{CO_2} = 1274t$ and black carbon emissions $M_{BC} = 0.403t$ per launch. The black carbon amplification factor is set as $\theta_{BC} = 5000$, consistent with conservative upper-bound estimates for stratospheric black carbon radiative forcing. Stratospheric water vapor injection is estimated as $M_{H_2O} = 400t$ per launch, with a stratospheric flight fraction $f_{strat} = 0.8$ and a scaling coefficient $\delta = 0.02$. The space elevator system, powered by electricity, is assumed to generate negligible direct atmospheric emissions during operation.^[9]

7.2 The Establishment of EIOD Model

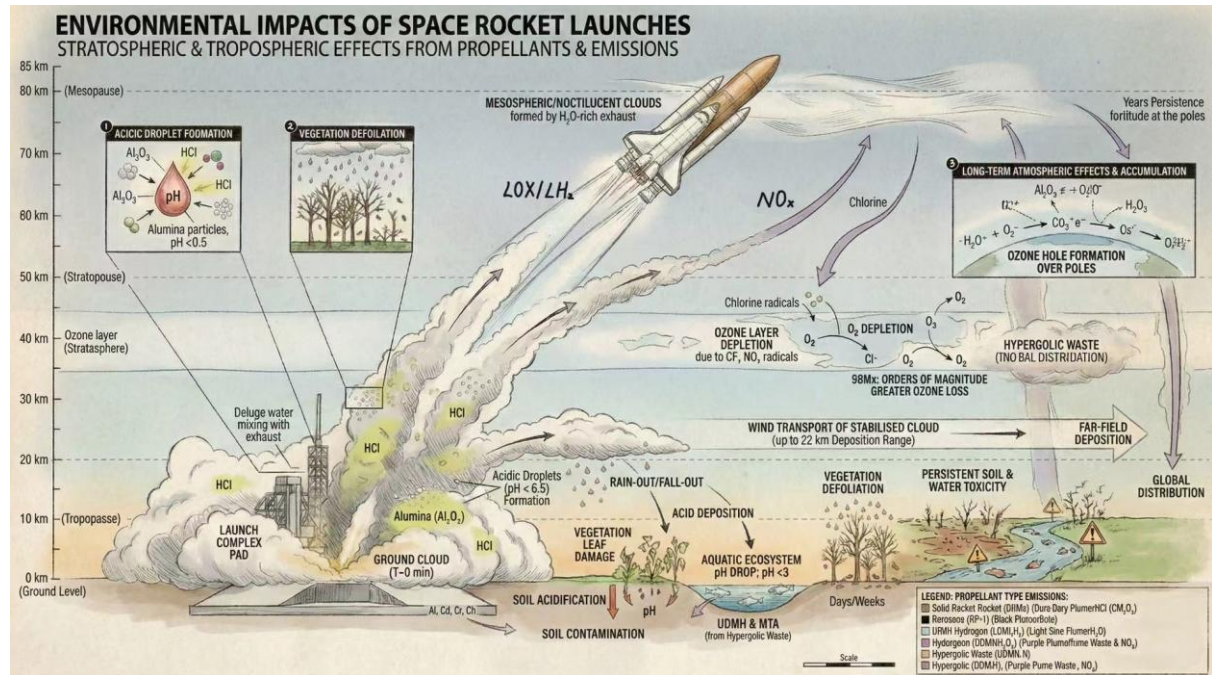


Figure 7: Impact on the Earth's environment for achieving the Moon Colony

To evaluate and compare the environmental consequences of alternative transportation strategies, an Environmental Impact Coefficient (EIC) is constructed to represent the equivalent surface CO_2 emissions associated with space transportation activities.

Rather than estimating absolute climate impacts, the EIC serves as a **relative metric**, enabling consistent comparison across different logistics scenarios under uncertainty. The model is grounded in the concept of radiative forcing equivalence commonly adopted in atmospheric physics and life-cycle assessment studies.

7.2.1 Definition of Environmental Impact Coefficient

For a single rocket launch, the EIC is defined as:

$$EIC = M_{CO_2} \cdot (1 + \lambda_{BC} + \lambda_{strat}) \quad (35)$$

Where M_{CO_2} denotes the mass of carbon dioxide directly emitted during launch, λ_{BC} represents the relative warming contribution of black carbon emissions, λ_{strat} captures the additional climatic impact associated with high-altitude emissions.

7.2.2 Black Carbon Enhancement Term

The black carbon enhancement factor is modeled as:

$$\lambda_{BC} = \theta_{BC} \cdot \frac{M_{BC}}{M_{CO_2}} \quad (36)$$

Where M_{BC} is the emitted black carbon mass and θ_{BC} is an amplification coefficient reflecting the elevated radiative forcing efficiency of black carbon in the upper atmosphere. Following existing literature, θ_{BC} is selected from a conservative mid-range of reported values.

7.2.3 Stratospheric Emission Adjustment

To account for combined effects of water vapor and nitrogen oxides injected into the stratosphere, a simplified correction term is introduced:

$$\lambda_{strat} = \delta \cdot f_{strat} \quad (37)$$

Where f_{strat} denotes the fraction of the flight trajectory occurring within or above the stratosphere, and δ is a scaling coefficient representing the aggregate radiative influence of high-altitude emissions.

7.3 The Solution of EIOD Model

The calculated MIS is 3309.384, with a marginal environmental impact per ton of

$$MIT = \frac{EIC}{M_{payload}} \quad (38)$$

Resulting in a total environmental impact level of

$$Z_3 = MIS \cdot Q_{Rocket} \quad (39)$$

By incorporating environmental impact minimization as a third objective, a three-objective Pareto-optimal transportation strategy is obtained. Under this strategy, 88.17% of supplies are transported via the space elevator, with a project duration of 89.12 years and a total cost of 141.05 trillion USD.

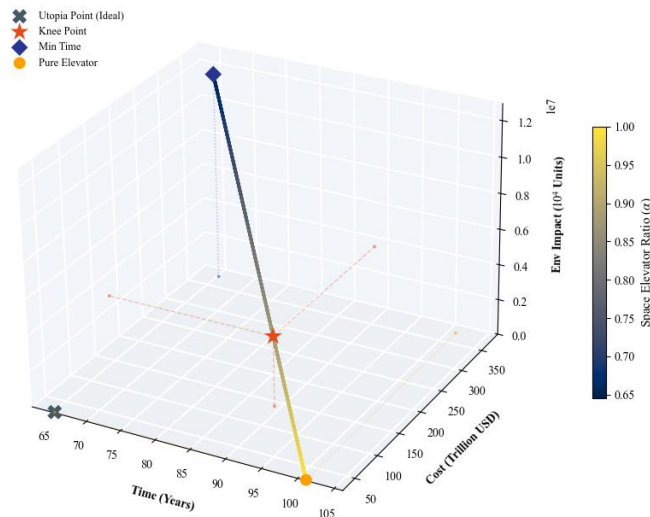


Figure 8: Distribution of the Pareto Frontier in the EIOD Model

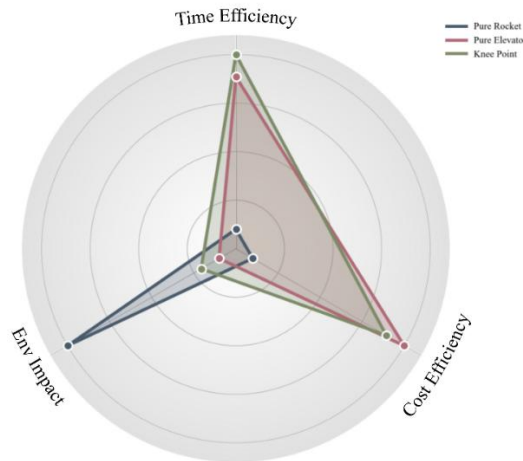


Figure 9: Three-Objective Radar Chart in the EIOD Model

8 Sensitivity Analysis

A sensitivity analysis is conducted on the unit transportation cost of rockets, which is reduced from the baseline value of 10,000 to 250, matching that of the space elevator. Results show that when the cost remains above 250, the Pareto-optimal solution is unchanged, with a stable transportation share assigned to the space elevator.

Once the cost falls below 250, the optimal solution structure shifts abruptly. The corresponding curve of space elevator usage ratio exhibits a near-vertical change below this threshold and remains nearly horizontal above it. This reveals a clear cost tipping point, confirming the piece-wise stability of the proposed model.

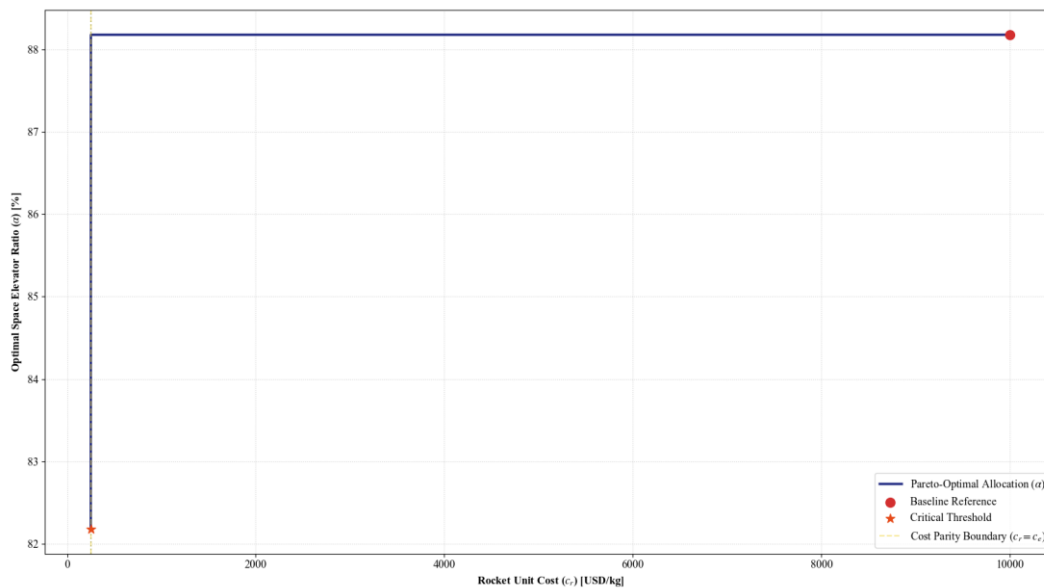


Figure 10: Impact of Reducing Rocket Cost on the EIOD Model

9 Model Evaluation and Further Discussion

9.1 Strengths

Our model demonstrates a systematic approach to multi-objective optimization. The cost-time Pareto frontier allows decision-makers to assess trade-offs across strategies rather than relying on a single solution. By integrating dynamic programming with capacity constraints, inventory and transportation lead times are handled as an asynchronous scheduling problem, enhancing realism. Incorporating non-ideal factors such as equipment failures and capacity fluctuations improves robustness. The paper links data and models closely, with clear visualizations showing how key variables influence strategy selection, reflecting rigorous analysis and interdisciplinary thinking. Overall, the approach balances theoretical rigor with practical applicability.

9.2 Weaknesses

The model has limitations. Linear cost assumptions ignore economies of scale, potentially overestimating total costs. Technology readiness is idealized, assuming full-capacity elevator

operation from the start, which may overstate early-stage strategy effectiveness.

9.3 Further Discussion

Future work may incorporate time-varying population growth, stochastic demand, and adaptive launch scheduling. Extending the model to include technological learning curves, multi-elevator networks, and policy constraints would further enhance its practical applicability and feasibility for large-scale space infrastructure planning.

10 Conclusion

Our work develops an integrated cost–time–environment optimization framework for large-scale space transportation systems. By combining dynamic programming, multi-objective Pareto analysis, and robustness considerations, the model systematically compares space elevators and conventional rockets under both ideal and non-ideal conditions. Results indicate that space elevators dominate long-term transportation efficiency, while rockets retain flexibility under short-term or failure-prone scenarios. The proposed framework provides a quantitative decision-making foundation for future extraterrestrial infrastructure planning.



Dear MCM Agency:

Establishing a Moon Colony for 100,000 residents is not merely an engineering task but a strategic systems decision balancing cost, time, reliability, and environmental impact. Based on a multi-objective optimization framework and Pareto-front analysis under both ideal and non-ideal conditions, we recommend a phased hybrid transportation strategy centered on the Space Elevator System.

During construction, exclusive reliance on a single transport mode is inefficient. Rocket-only delivery incurs prohibitive costs, while sole dependence on the Space Elevator—comprising three Galactic Harbours, each supported by Earth Ports, dual 100,000 km graphene tethers, and super-GEO anchors—leads to excessively long durations. Incorporating operational uncertainties such as elevator downtime, tether dynamics, and rocket failures, the Pareto-optimal knee-point strategy allocates approximately 88.17% of materials to the Space Elevator and the remainder to rockets.

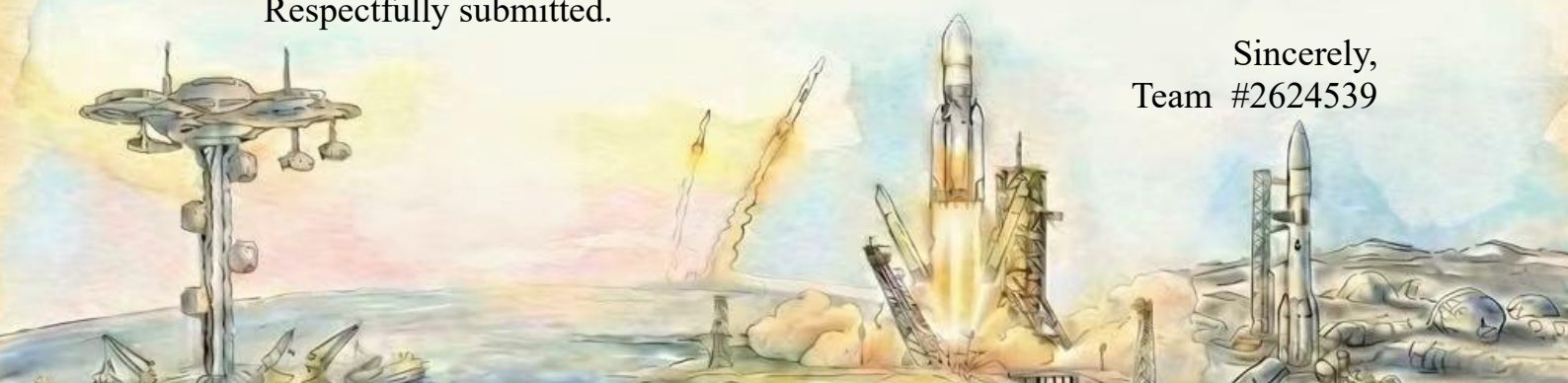
This approach completes primary construction in 89.12 years at a total cost of 141.05 trillion USD, minimizing distance to the utopian point on the Pareto frontier. Compared with time- or cost-minimizing extremes, it offers a robust and feasible solution under realistic constraints.

For long-term colony operations, particularly annual water supply, we recommend extending this hybrid framework into a steady-state logistics model. High-frequency bulk transport should use the Space Elevator to minimize marginal cost and environmental impact, while rockets provide responsive backup for contingencies and peak demand. This significantly reduces cumulative atmospheric emissions and aligns with sustainability goals.

In summary, prioritizing the Space Elevator while retaining strategic rocket support provides a technically sound, economically viable, resilient, and environmentally responsible pathway toward sustained lunar habitation.

Respectfully submitted.

Sincerely,
Team #2624539



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Use of AI Tools in COMAP Contests

1.OpenAI ChatGPT(GPT-5.2)

Question

How to revise informal or engineering-style expressions into academically appropriate language consistent with scholarly writing standards.

Response

Suggestions were given for formal rephrasing, including noun-based expressions, passive constructions, and terminology refinement to improve academic tone and clarity.

Content Adopted

Language in the background description, model assumptions, and result interpretation sections was revised to enhance formality and consistency with academic writing conventions.

2.OpenAI ChatGPT(GPT-5.2)

Question

How to adjust figure color schemes to improve readability and visual coherence while remaining appropriate for academic publications.

Response

Recommendations were given on using low-saturation, high-contrast color palettes, with optional thematic adjustments to better match the space-related context of the study.

Content Adopted

Color schemes of cost–time trade-off plots and inventory evolution figures were modified to improve clarity and overall visual quality.

3.OpenAI ChatGPT(GPT-5.2)

Question

How to optimize legend placement, annotation positions, and font settings to avoid visual overlap and improve presentation quality.

Response

Specific suggestions were given regarding legend positioning, font selection (e.g., Times New Roman), and annotation spacing to enhance figure professionalism.

Content Adopted

Figures were reformatted accordingly, resulting in clearer layouts and improved readability without altering the underlying data or results.

4.Google Gemini(Gemini 3)

Question

How to adjust the aspect ratio of the existing hand-drawn space illustration to an ultra-wide 32:9 ratio.

Response

Extension of the background horizontally to both sides while maintaining the original

style, supplemented with starfields and minor celestial elements.

Content Adopted

The generated ultra-wide background image, used as header decoration for the presentation slides.

5.Google Gemini(Gemini 3)**Question**

How to modify the astronaut's boots to create an integrated look with the spacesuit while maintaining the hand-drawn texture.

Response

Re-rendering of the lower body joints to blend the lines of the boots and suit, maintaining the original watercolor/sketch texture.

Content Adopted

The modified astronaut figure, used as a visual asset in the "Application Scenarios" section of the paper.